

ECE 2201 Microelectronic Circuits I

Lab #2: Diode Applications

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EQUIPMENT

- ECE Lab Kit
- DC Power Supply
- 12VAC Transformer
- DVM
- Function Generator
- Oscilloscope

Abstract

This experiment explores the behavior of diode-based rectifier circuits and reveals how the addition of capacitive filtering can improve the quality of DC output. We started with the half and full wave rectifiers regarding the 1N4004 diodes. Through that we explored how diode behaviors influence waveforms and gained an understanding of how capacitive filtering can reduce the ripple. Additionally, this lab gears towards regulated supplies by utilizing a Zener diode voltage regulator and a LM7805 linear regulator, which both provide voltage stability under various loads. This experiment examines the measured waveforms, the transformer operation, ripple behavior, and the dropout limits of the voltage regulators.

Introduction

The purpose of Lab 2 – Diode Applications is to gain an understanding of how AC signals can transform into unstable DC voltages through rectification, filtering, and regulation. This lab starts with the basic half wave and full wave rectifier circuits, where the diode behavior can determine the parts of the AC waveform that reach the load. The effects of adding the filter capacitors are then studied and analyzed showing how capacitor size can directly affect the ripple amplitude and smoothness of the DC output. This lab then goes further and investigates voltage regulation through the methods of a Zener diode-based regulator and a LM7805 linear regulator. The circuits reveal how a regulated power supply can maintain a near constant output voltage even if the load resistance is changing. Ultimately, this lab reveals how unregulated and regulated supplies have their differences and shows how each stage can enhance the stability and quality of DC output.

W1

L1 Circuit Description

The rectifier circuit for W1 is built with the 12VAC transformer, a 1N4004 diode that is in series and a $1k\Omega$ load resistor that is connected to the ground. This diode is oriented to be forward biased during the positive half-cycle which allows the current to flow at only times when the transformer is having positive output. The oscilloscope was able to measure VAC1 directly through the transformer and then V_L across the load resistor. This configuration of the circuit produces single polarity output and blocks all the negative half cycles.

W1-1.

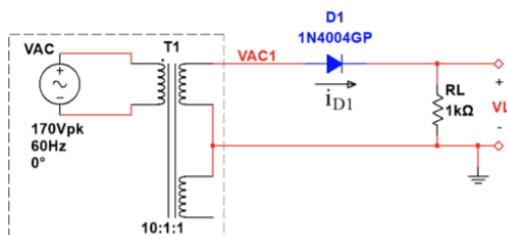
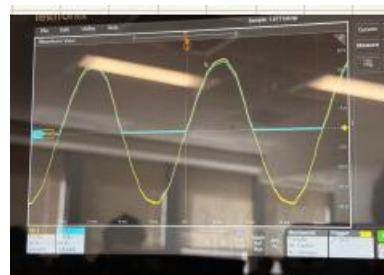


Figure 1.1 Half-Wave Rectifier Circuit (no filter cap)



Waveform of Figure 1.1

In between the waves, there was a 0.7V difference at the peak in positive range. V_L will not produce any output in the negative range of VAC1.

W2

L2 Circuit Description

The rectifier from W1 was modified to have a $10\mu\text{F}$ capacitor that is in parallel with the $1\text{k}\Omega$ load resistor. The capacitor has its positive end connected to the node of the diode's output and its negative connected to the ground. This allows for the capacitor to charge to the peak of the rectified waveform during conduction and then slowly discharge through the load during the non-conduction periods. Channel one in the oscilloscope was measuring the transformer output (VAC1) and channel two was measuring the output voltage across the capacitor and load resistor (R_L).

W2-1.

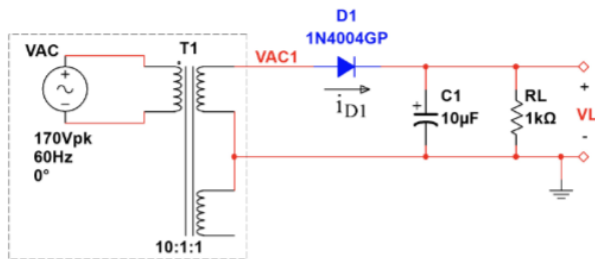


Figure 2.1 Half-Wave Rectifier Circuit (with filter cap)

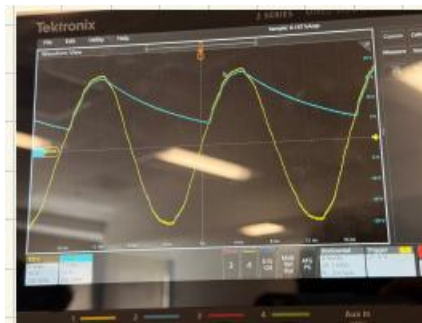


Figure 2.2 Waveform of Figure 2.1 with $10\mu\text{F}$

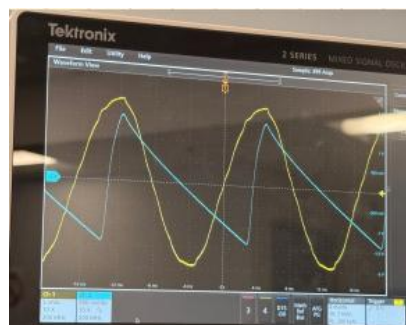
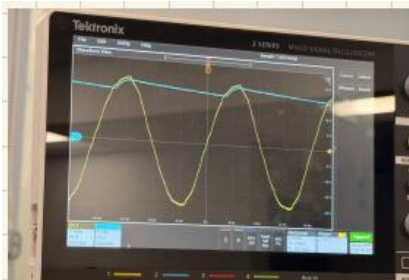


Figure 2.3 Waveform of Figure 2.1 with 100 μ F

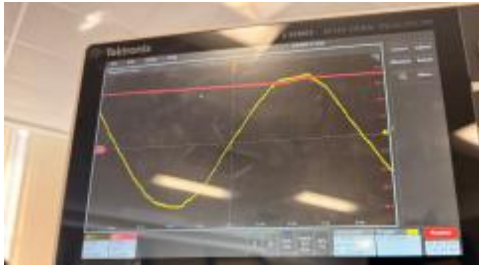


Figure 2.5 Waveform of Figure 2.1 with 330 μ F

Figure 2.4 AC coupling Waveform of Figure 2.1 with 10 μ F

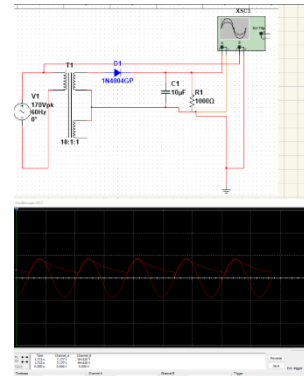


Figure 2.6 Waveform simulation of Figure 2.1 with 10 μ F

W2-2.

The ripple voltage in a half-wave rectifier with a filter capacitor is determined through how quickly the capacitor can discharge the load between the conduction peaks. The theoretical relationship is determined through the formula: $V_{RPP} = V_P * \frac{1}{FRC}$. This shows that ripple decreases when capacitance increases and will increase when the load resistance is smaller. With the fixed load resistance of 1k Ω , measurements were made for several capacitance values. In figure 2.2 there is a 10 μ F capacitor that charges rapidly with a steep upward slope as well as a large ripple amplitude. Then, in Figure 2.3, a 100 μ F is used that decreases the ripple significantly while voltage staying near the peak for a larger amount of the cycle. Also, as shown Figure 2.6, with an image of the simulated circuit as well as the result, its waveform has a nearly identical shape as the experimental result in Figure 2.2.

W2-3.

To attempt to reduce the ripple to less than 1V_{pk-pk}, the ripple formula stated above in W2-2 was used. After plugging in the known values into the equation, $1 = 20 * \frac{1}{60 * 1000 * C}$, the calculation predicts that a capacitance of 333.333 μ F is needed to bring the ripple under 1V_{pk-pk}. Since the ECE lab kit only provides standard capacitor values, 330 μ F was used. After testing in the circuit, the measured waveform confirmed the prediction with the ripple voltage dropping to about 1V_{pk-pk} and the output voltage also staying very close to the peak between the conduction intervals. This matches the theory that the larger capacitance will increase the time constant (RC) resulting in the capacitor discharging more slowly through the 1k load and dramatically decreasing the ripple.

W3

L3 Circuit Description

The circuit used in W3 is a full-wave rectifier circuit that is built with a center tapped transformer with two diodes. Each half of the transformer's secondary has alterations of polarity resulting in one diode

conducting in the positive half-cycle and the other to conduct in the negative half-cycle. This results in two positive pulses per one AC cycle causing a full wave rectified to 120Hz. With no filter capacitor being used the output voltage will drop to 0V between peaks.

W3-1.

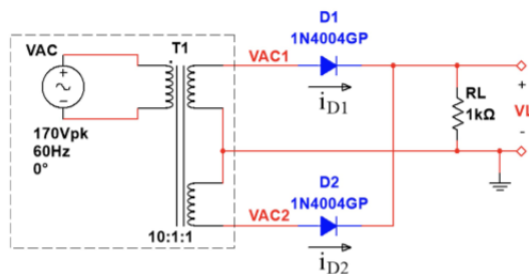


Figure 3.1 Full-Wave Rectifier Circuit (no filter cap)

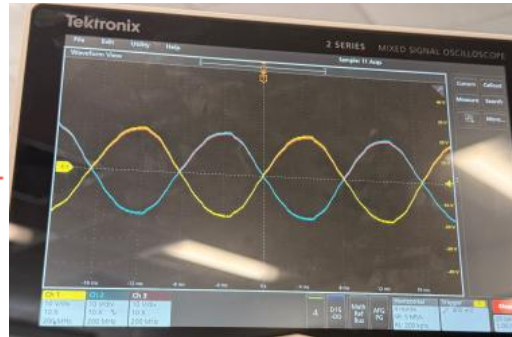


Figure 3.2 Waveform of Figure 3.1 with channel 1 being VAC1 and Channel two being VAC2

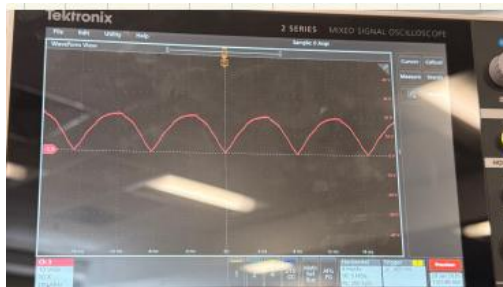


Figure 3.3 Waveform of Figure 3.1 with channel 3 being V_L

Based on the figures above, VAC1 and VAC2 are both sine wave with same amplitude and frequency but are 180 degrees out of phase with one waveform being positive while the other is negative. The full wave rectifier output produces two positive pulses per cycle which doubles the ripple frequency when compared to the half-wave rectifier. Differently than the half-wave case, the voltage does not skip any cycle and is always conducting but still will drop to 0V between peaks because there is no capacitor in this configuration.

W4.

L4 Circuit Description

The full-wave rectifier circuit from W3 was updated with the addition of a filter capacitor in parallel with the 1kΩ load resistor. The positive end of the capacitor was connected to the rectifier's output node, and the negative end was to the ground. This configuration allows the capacitor to charge to its peak of full

wave rectified signal and then slowly discharge through the load during the periods of changing pulses. Channel one of the oscilloscope measuring VAC1, channel two was measuring VAC2, and channel three was measuring the output across the capacitor and load resistor.

W4-1.

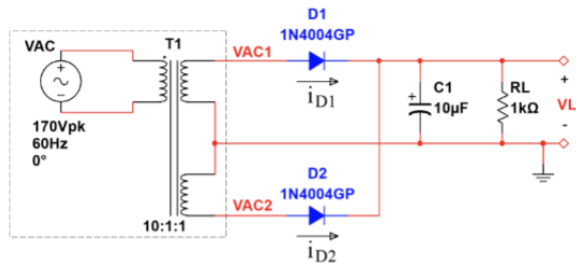


Figure 4.1 Full-Wave Rectifier Circuit (with filter cap)

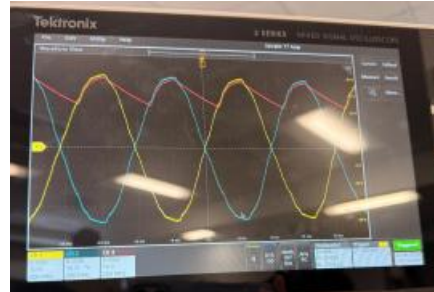


Figure 4.2 Waveform of Figure 4.1 with 10µF

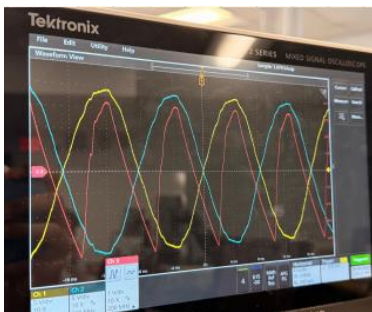


Figure 4.3 AC coupling Waveform of Figure 4.1 with 10µF

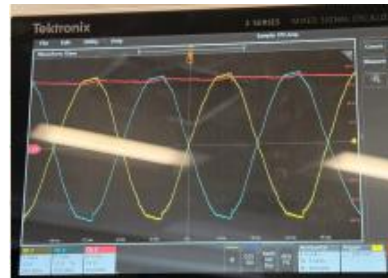


Figure 4.4 Waveform of Figure 4.1 with 200µF

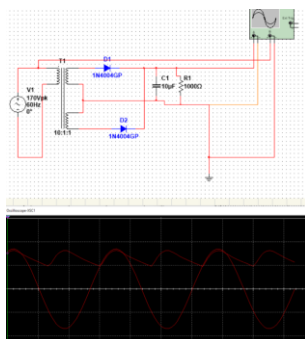


Figure 4.5 Waveform simulation of Figure 4.1 with 10µF

W4-2.

For a full-wave capacitor, the ripple voltage is the equations: $V_{RPP} = V_P * \frac{1}{2FRC}$. This shows that ripples have an inverse relationship with both capacitance and load resistance. Larger capacitors should be able to discharge slower and reduce the ripple and larger load resistors will also reduce ripple by less current being drawn from the capacitor. Also, as shown in Figure 4.5, with an image of the simulated circuit as well as the result, its waveform has a nearly identical shape as the experimental result in Figure 4.2.

W4-3.

In the design challenge, a ripple peak to peak of 1V or less is needed. Using the equation from W4-2 and plugging in the known values, $1 = 20 * \frac{1}{2 * 60 * 1000 * C}$, it was calculated that the capacitance needs to be 166.67 μ F to achieve a 1V_{pk-pk} ripple. Due to the limitations of the ECE lab kit, two 100 μ F capacitors in parallel were used yielding 200 μ F which is approximately the needed 166.67 μ F. The oscilloscope measurements revealed that the ripple was indeed less the 10 μ F case and had approximately a 1V_{pk-pk}. Overall, the measured experimental results as well as the simulated results both support the theoretical prediction that capacitance of 166.67 μ F or high yield ripple less than 1V_{pk-pk}.

W4-4.

The full-wave rectifier gives two changing pulses per one cycle which greatly reduces the ripple with the same load and capacitance applied in comparison to the half-wave circuit. This is because the capacitor is refreshed twice as much and the full-wave design results in a higher more stable DC output. The main advantage is increased efficiency and smoother voltage when the half-wave rectifier is simpler but ends up suffering with larger ripple and worsened utilization of the AC waveform. Ultimately, the full-wave circuit is better for power supply application with its only con being more circuit complexity.

W5.

L5 Circuit Description

The circuit in L5 is a full-wave rectifier built with a center-tapped transformer and two diodes. Each half of the transformer's secondary alternates the polarity. This allows one diode to conduct in the positive half-cycle and the other to conduct during the negative half-cycle, allowing two positive pulses per cycle with a 120Hz full-wave rectified signal. It is updated from the circuit in L4, as we added two 510 Ω resistors in parallel to fashion a 250 Ω resistor in series. Next, we swapped out the 10 μ F filter capacitor for a 100 μ F filter capacitor. We then updated the circuit through the addition of a Zener diode parallel with the filter capacitor and 1k Ω resistor. The Zener diode voltage regulator will be used to achieve an even lower ripple and a better approximation to the ideal of a constant DC voltage.

W5.1

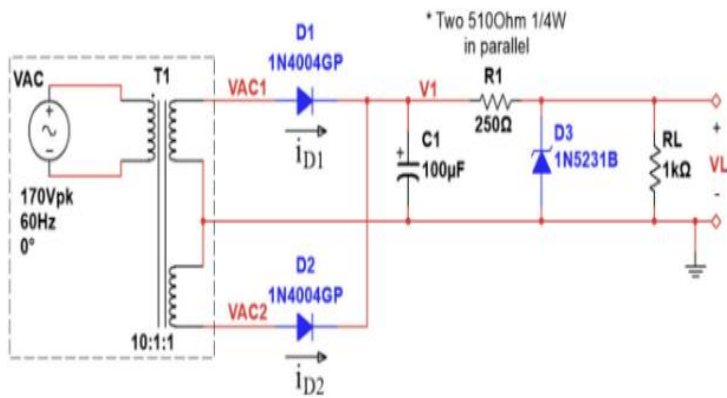


Figure 5.1 Full-Wave Rectifier Circuit (with cap and Zener regulator)

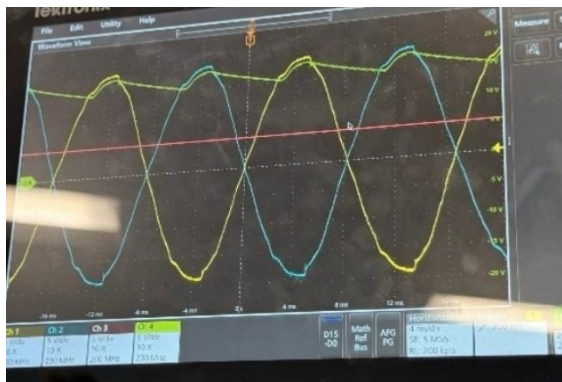


Figure 5.2 Waveform of Figure 5.1 with 100μF and Zener

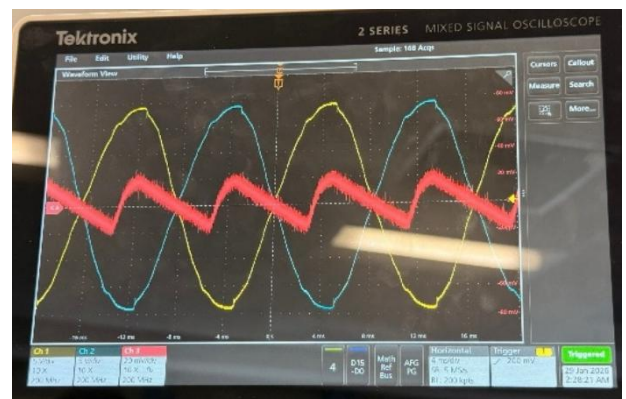


Figure 5.3 AC coupling Waveform of Figure 5.1 with 100μF and Zener

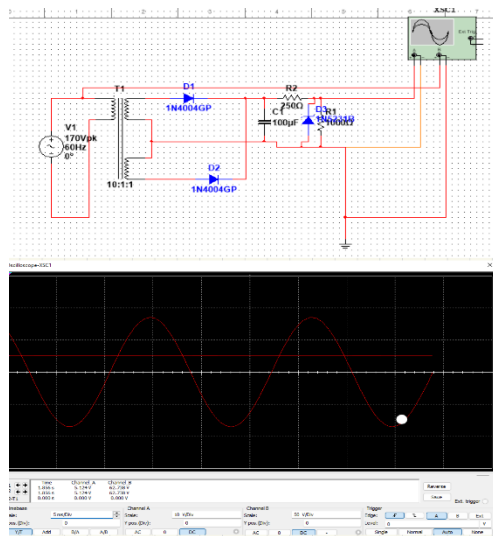


Figure 5.4 Simulation and graph of Figure 5.1 showing VL and VAC1 being on the graph

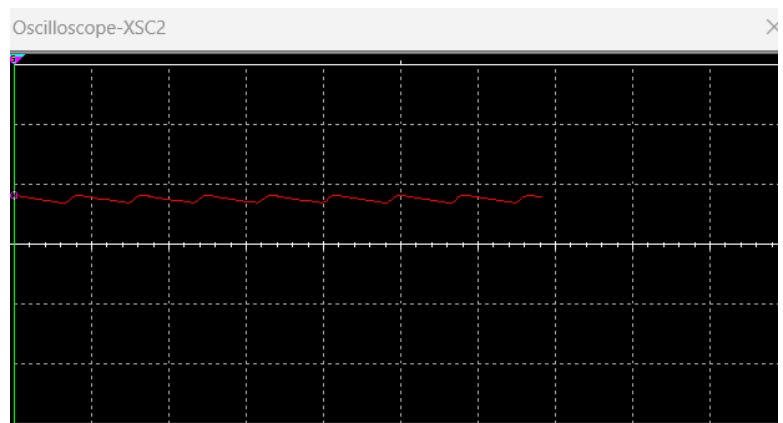


Figure 5.5 Continuation of Figure 5.1 showing graph of V1

W5.2

The experimental waveforms taken during lab are very similar to the Multisim simulation with VAC1 and VAC2 appear as the expected 60Hz sine waves with the rectified capacitor voltage of V1 showing similar ripple levels as in simulation and in lab. For the load voltage the output appears as a virtually constant ~5.1 V both in lab and in the simulation. This proves that the Zener is operating in a breakdown and behaving as it was predicted.

W5.3

$$\begin{aligned}
 V_L &\approx 5.1V \\
 P_{Load} &= \frac{V_L^2}{R_L} = \frac{5.1^2}{100} = 0.02601W \\
 \frac{I_1 - V_L}{250} \cdot V_1 &= P_{Zener} \\
 I_1 &= \frac{V_1 - V_L}{R_1} = \frac{17 - 5.1}{250} = 0.0476 \\
 P_{Zener} &= V_1 \cdot I_1 = 2048 \cdot 17 = 0.81 \\
 \eta &= \frac{0.0260}{0.81} \cdot 100\% = 3.2\%
 \end{aligned}$$

Figure 5.5 work to calculate experimental

efficiency of Figure 5.1

Without the Zener regulator, the output ripple is larger as its the capacitor discharging across RL. With the Zener regulator, the voltage is held at around 5.1V and the measured ripple shown at VL is much smaller than what is seen in L4.3

The Zener regulator was not efficient in this circuit. It had an efficiency of 3.1% due to the amount of excess voltage lost to heat.

$$\eta = \frac{P_{Load}}{P_{Source}} * 100, \eta = 3.1\%$$

W6

Circuit Description

The circuit in L6 is a full-wave rectifier built with a center-tapped transformer and two diodes. Each half of the transformer's secondary alternates the polarity. This allows one diode to conduct in the positive half-cycle and the other to conduct during the negative half-cycle, allowing two positive pulses per cycle with a 120Hz full-wave rectified signal. It is updated from the circuit in L5, as we replaced the Zener diode with a LM7805 linear regulator IC. We also removed the two 510Ω resistors and replaced them with a 10Ω "sense resistor" to measure current into the LM7805. This way we can supply a steady source of voltage to power loads.

W6.1

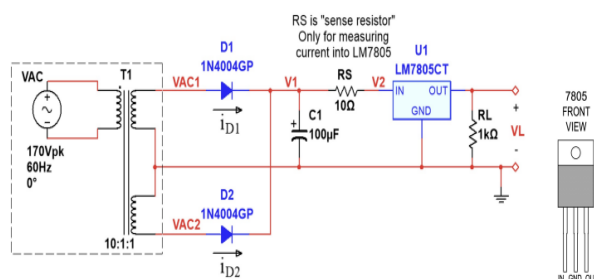


Figure 6.1 Full-Wave Rectifier Circuit (with cap and LM7805 linear regulator)

Figure 6.2 Waveform of Figure 6.1 with and LM7805



Figure 6.3 Waveform of Figure 6.1 with and LM7805

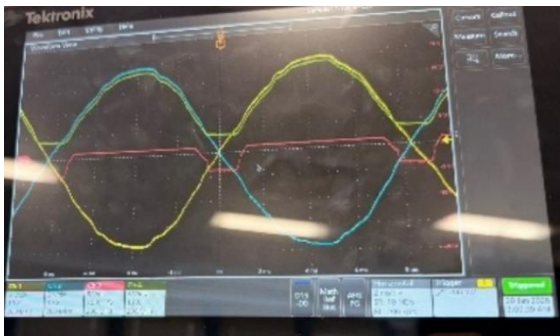


Figure 6.4 Part 1 Waveform of Figure 6.1 with and LM7805



Figure 6.5 Part 2 Waveform of Figure 6.1 with and LM7805

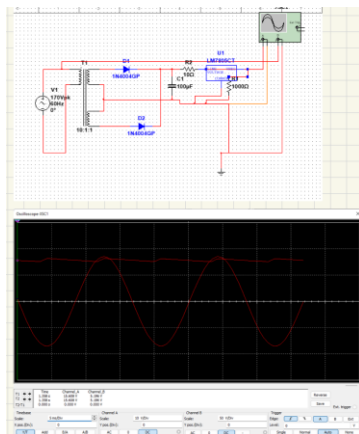


Figure 6.6 simulation of Figure 6.1

W6.2

Compare calculated values to measurement and simulation

The measured waveforms in the LM7805 circuit match the Multisim pretty well with the VAC1 and VAC2 are the expected 60Hz sinew waves. The v1 was the full wave rectified waveform with a small

ripple just like the simulation and the LM7805 can regulate it to keep it at about 5V DC. It confirms that the output is properly regulated at 5V with R_L at $1k\Omega$ and I_L being 5mA.

W6.3

The LM7805 was able to prove to be pretty effective at maintaining a constant 5V output in L6-3, L6-4, L6-5. There is minimal ripple showing that the regulator has strong load regulation. When the load gets larger, the input power is drawn and the sense resistor increases which is expected but the regulated output remains unchanged. When the load decreases the regulator smoothly reduces its current drawn and keeps the output at 5V which shows it has good stability.

W6.4

Handwritten calculations on graph paper:

$$I_3 = \frac{(V_1 - V_2) \times 10^3}{R_3} = \frac{18 - 17.8}{10} = 0.02 \text{ A}$$

$$P_{out} \approx 0.025 \text{ W}$$

$$P_{dis} = I_3 \times (V_1 - V_2) = 0.02 \times 18 = 0.36 \text{ W}$$

$$\frac{0.025}{0.36} \approx 6.94\%$$

$$P_{out} \approx \frac{5^2}{100} = 0.25 \text{ W}$$

$$V_{R3} = I_3 \times 10 \text{ } =$$

$$I_3 = \frac{V_1 - V_2}{10} = \frac{18 - 17.3}{10} = 0.07 \text{ A}$$

$$P_{dis} = 1.2 \text{ W}$$

$$\frac{0.25}{1.28} \approx 19.8\%$$

Figure 6.7 and 6.8 Work to calculate efficiency

The measured efficiency of the LM7805 regulator was 6.94% for L6-2 with the $1k\Omega$ load resistor and 19.8% for L6-4 for the 100Ω load resistor. This reveals that the regulator is more noticeable when under a heavier load and since the larger fraction of input power is delivered instead of being dissipated in the regulator and resistor. In comparison to the Zener from earlier it had lower efficiency, there was an increase in efficiency because in the Zener diode it continuously was burning the extra current which reveals that the LM7805 is more efficient and provides better regulation.

W6.5

In this design challenge, we decided to use the 10uF and the 100uF capacitors, bringing us two different graphs shown in figure 6.4 and 6.5. To determine the required capacitance, we used the ripple voltage equation for a full wave rectifier. $I_{Load} = \frac{5V}{51\Omega}$ is about 0.098A.

Because the LM7805 requires approximately 2V of dropout voltage to remain in regulation, we needed to make the capacitor large enough to prevent the voltage dropping below 7V.

The 100uF capacitor was sufficient as it was the minimum required to fulfill the circuit's needs and reduce the ripple, allowing the LM7805 to maintain a steady 5V output. The 10uF capacitor failed as it could not regulate the ripple enough. This verified out theoretical prediction.

Discussion

This lab showed how each step in building a DC power supply helps clean up and stabilize the output voltage. Our measurements for the half-wave, full-wave, and filtered rectifiers lined up well with both the theory and the Multisim results, especially in how the ripple changed with different capacitor values. When we added voltage regulation, the differences became really clear with the Zener diode held the voltage around 5 V but ended up being very inefficient, while the LM7805 kept the output almost perfectly at 5 V and handled load changes much better. The efficiency numbers of 3.1% for the Zener and 6.94% to 19.8% for the LM7805, it shows that linear regulators work more efficiently when the load draws more current. Overall, the lab helped us see how rectifiers, filters, and regulators all work together to create a steady DC power supply.

Conclusion

The purpose of this laboratory experiment was to investigate the operation and performance of several DC power supply configurations using half-wave and full-wave rectifiers, capacitive filtering, and voltage regulation techniques. We found the results demonstrated adding a filter capacitor significantly reduces ripple voltage by storing charge and releasing it between peaks of the rectified waveform. Additionally, the full-wave rectifier proved to be better than the half-wave rectifier as it recharges the capacitor twice per cycle, resulting in a smoother DC output. Through introducing voltage regulators, it further improved the circuit performance by maintaining a nearly constant output voltage despite fluctuations in the input. The Zener regulator successfully limited the output to about 5V but had low efficiency due to power dissipation as heat loss in the series resistor and diode. The LM7805 linear regulator provided a more stable and reliable output with improved load regulation, though it also exhibited reduced efficiency because excess voltage was dissipated as heat. The design challenges confirmed the theoretical relationship between ripple voltage, capacitance, and load current. Increasing capacitance reduced ripple and helped maintain the minimum input voltage required for proper regulator operation, particularly when accounting for the LM7805 dropout voltage. Overall, the measured experimental results closely matched theoretical predictions and simulations, validating the design methods used to create a stable DC power supply.